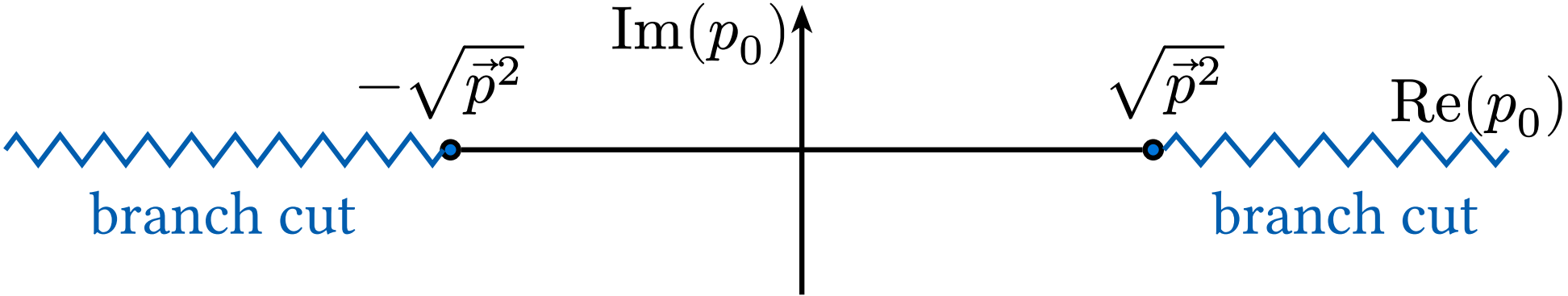
$\mathrm{Im}(p_0)$
$-\sqrt{\vec{p}^2}$
$\sqrt{\vec{p}^2}$
$\mathrm{Re}(p_0)$
branch cut
branch cut